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Titanium and nitrogen co-doped porous carbon
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A novel titanium, nitrogen co-doped carbon material was designed via a one-step solvothermal reaction at a moderate temperature. Characterization by scanning electron microscopy and transmission electron microscopy illustrates that carbon materials without titanium doping possess a morphology of porous flakes, whereas the inclusion of titanium results in the deposition/intercalation of island-shaped TiO₂ nanoparticles on these carbon flakes. Both the volumetric energy density and the cycling stability are significantly improved by titanium doping. Electrochemical measurements show that the introduction of titanium leads to a high volumetric capacitance of 285.5 F cm⁻³ at 0.5 A g⁻¹ current density. Electrodes prepared with the new materials also exhibit excellent cycling stability, where there is no capacitance loss after 40 000 cycles in the 6 M KOH electrolyte at a high charge/discharge current of 30 A g⁻¹. The volumetric energy density of the as-obtained symmetrical supercapacitor reaches 11.99 W h L⁻¹, which is competitive to that of lithium thin-film batteries (1–10 W h L⁻¹).

1. Introduction

Supercapacitors (SCs) have been considered as one of the most sustainable energy-storage techniques for advancing the Internet-of-Everything, portable electronics, electric vehicles and smart grids, owing to their high power density and fast charging/discharging rates.^{1–14} However, the low energy density of SCs severely impedes their practical applications. In order to satisfy the ever-growing energy demands, highly pseudocapacitive materials have been targeted for high-performance SCs, such as RuO₂,¹⁵ Ni(OH)₂,¹⁶ Fe₃O₄,¹⁷ TiO₂,¹⁸ V₂O₅,¹⁹ etc., which generally provide higher electrochemical capacitance and energy density than those electrode materials applied in traditional electrical double-layer capacitors because there are extra surface redox reactions involved in pseudo-capacitive SCs.

Among them, TiO₂ has been widely studied because of its low cost, abundance, environmental friendliness, high capacitance and high-current-rate tolerance for large-scale energy storage.^{20–28} However, the low electronic conductivity and poor electrochemical stability critically deteriorate the electrochemical performance of TiO₂-based SCs. Early studies suggested

that the direct growth or deposition of TiO₂ nanomorphologies on conductive materials would not be able to improve the electronic conductivity and capacitive performance of TiO₂ materials, due to their intrinsic semiconductor nature and barely non-intimate contacts to the conductive materials. Later on, many researchers tried to confine TiO₂ nanomaterials within the conducting matrix to enhance the overall conductivity, yet, the advantages of high capacitance derived from surface redox reactions on TiO₂ were largely diminished by the low rate of ion diffusion in non-desired pores and channels.^{27,28} Therefore, there is an urgent need for the design of delicate nanostructures between TiO₂ and conductive materials.

Herein, we report a one-step large-scale synthesis of high-density ($\rho_{\text{pack}} = 1.10 \text{ g cm}^{-3}$) Ti and N co-doped carbon flakes. Notably, the *in situ*-polymerization with the introduction of small amounts of transition metal Ti resulted in a unique morphology of TiO₂ nanoparticles embedded on the surface of carbon flakes, which renders the electrode materials with improved conductivity and efficient faradaic reactions. The as-prepared supercapacitor delivered a high gravimetric capacitance of 259.5 F g⁻¹ and a remarkable volumetric capacitance of 285.5 F cm⁻³. Moreover, electrodes also exhibit excellent cycling stability, where there is no capacitance loss after 40 000 cycles in a 6 M KOH aqueous electrolyte at a high charge/discharge current of 30 A g⁻¹. The calculated volumetric energy density in aqueous solution is up to 11.99 W h L⁻¹, which is competitive to that of lithium thin-film batteries (1–10 W h L⁻¹).

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2. Experimental

2.1. Synthesis of N-doped carbon (NOCIC)

The NOCIC was prepared by adding 24.0 mL of pyridine and 6.0 mL of hexachlorobutadiene to a 50 mL steel autoclave coated with Teflon, in which pyridine and hexachlorobutadiene act as the source of nitrogen and chlorine atoms. The sealed autoclave reactor was then placed in an oven for 24 h at 200 °C. After the reactor was cooled down to room temperature naturally, black solid products were collected and stored in a desiccator. To remove the possible residues of pyridine (boiling temperature of 115 °C) and hexachlorobutadiene (boiling temperature of 225 °C), the solid products were dried at 350 °C for 2 h and then washed several times with deionized water and dried again in a vacuum at 80 °C. For the thermal treatment, the above synthesized solid was placed in a tubular oven heated to a pre-determined temperature (*e.g.*, 250, 350, 450 and 550 °C) at a rate of 10 °C min⁻¹ under the protection of N₂ for 2 h. The flow of N₂ was kept at 35 mL min⁻¹. The materials processed at 350 °C are denoted as NOCIC. Our electrochemical experiments illustrate that materials thermally treated at 350 °C exhibit the best capacitance performance.

2.2. Synthesis of Ti, N-codoped carbon (TiNOCIC)

To synthesize the TiNOCIC, 24.0 mL of pyridine, 6.0 mL of hexachlorobutadiene and 400 mg of titanium nanoparticles were added to a 50 mL steel autoclave coated with Teflon. The sealed autoclave reactor was then placed in an oven for 6 h at 200 °C. After the reactor was cooled down to room temperature naturally, black solid products were collected and stored in a desiccator. For the thermal treatment, the above synthesized solid was placed in a tubular oven heated to a pre-determined temperature (*e.g.*, 250, 350, 450, or 550 °C) at a rate of 10 °C min⁻¹ under the protection of N₂ for 2 h, which are denoted as TiNOCIC (best electrochemical performance was obtained at 350 °C). The flow rate of N₂ was set as 35 mL min⁻¹. Finally, the thermally treated carbon materials were placed in a ball mill to undergo high energy ball milling for 1 min. This step is to make the particles of the synthesized carbon materials smaller.

2.3. Material characterization

Scanning electron microscopy (SEM) images were obtained using a field emission Nova Nanosem 200 system operated at an acceleration voltage of 15 kV. Both transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) were performed using a JEOL-3010 instrument. Energy-dispersive X-ray spectroscopy (EDS) analysis of the solid products was also carried out with the JEOL-3010 transmission electron microscopy. X-ray photoelectron spectroscopy (XPS) was performed on a spectrometer from Kratos Axis UltraDLD, using Mono Al K α radiation at a power of 120 W (8 mA, 15 kV). All binding energies were referenced to the C 1s peak (284.6 eV) arising from adventitious carbon. Low angle X-ray diffraction was performed on a Bruker D8 powder X-ray diffractometer using Cu K α radiation (λ = 0.15406 nm), where the data were collected in the 2θ range of 20–80° at a step size of 0.02°. Nitrogen adsorption-desorption isotherms were measured at

77 K using a Micromeritics Tristar 3000 system using vacuum-degassed samples (300 °C for at least 5 h). The isotherms were used to calculate (a) the specific surface area by the Brunauer–Emmett–Teller (BET) method and (b) the pore volume and pore size by the Barrett–Joyner–Halenda (BJH) method. Raman spectra of the samples were collected using a Renishaw in Via Raman microscope equipped with a laser of wavelength 532 nm. Weight fractions of the solid products were determined by thermogravimetric analysis (TGA; TA Instrument PerkinElmer Clarus). Determination of chlorine content in the samples was performed with a sulfur chlorine analyzer (THA-200, Taizhou, Jiangsu province). Conductivity data were measured by using a conductivity meter (DDS-11A_r). Densitometry was performed using a TD-1200 true density tester.

2.4. Electrochemical measurements

To fabricate electrodes, active materials (TiNOCIC, NOCIC, or as stated), PTFE, and acetylene black were mixed in a mortar according to a mass ratio of 8 : 1 : 1. Typically, 3 to 5 mg of the above mixture was spread on a nickel foam or a steel net, and dried in a vacuum oven at a temperature of 100 °C for 6 h. The nickel foam or steel net was subsequently compressed under a pressure of 200 kg cm⁻² for 1 min. Electrochemical experiments were performed in several configurations and electrolytes. In a three-electrode cell with alkaline electrolyte (6 M KOH), the Hg/HgO electrode was used as the reference electrode and Pt wire as the counter electrodes and cyclic voltammetry (CV) was performed between –1.1 and 0 V. Cyclic voltammetry and charge-discharge (CC) tests were conducted using an electrochemical analyzer, CHI760D (Chenhua, Shanghai), under ambient conditions. Electrochemical impedance spectroscopy (EIS) was performed with AUTOLAB, from 10 mHz to 100 kHz at the open circuit voltage with an alternate current amplitude of 5 mV. The gravimetric and volumetric capacitances of the electrode were calculated using galvanostatic charge/discharge curves according to the following equations:

$$C_s \text{ (F g}^{-1}\text{)} = \frac{I \times \Delta t}{m \times \Delta V} \quad (1)$$

$$C_{vol} \text{ (F cm}^{-3}\text{)} = C_s \times \rho \quad (2)$$

$$C' \text{ (F)} = \frac{1}{2\pi f Z''} \quad (3)$$

where C_s is the gravimetric capacitance of the electrode, C_{vol} is the volumetric capacitance of the electrode, and C' is the frequency-dependent imaginary capacitance. I (amperes) is the discharge current, Δt (seconds) is the discharge time, m (g) is the mass of the electrode materials, ΔV is the potential window during the discharge process after the IR drop, ρ is the density of the electrode, f is the frequency, and Z'' is the capacitive reactance resistance.

The gravimetric capacitances of one electrode of the symmetric capacitor device were calculated using galvanostatic charge/discharge curves according to the following equations:

$$C_s \text{ (F g}^{-1}\text{)} = \frac{2I \times \Delta t}{m \times \Delta V} \quad (4)$$

where C_s is the gravimetric capacitance of the single electrode, I (in ampere) is the discharge current, Δt (in second) is the discharge time, m (in g) is the mass of the single electrode, and ΔV is the potential window during the discharge process after the IR drop. The specific capacitance of the supercapacitor cell was evaluated according to the following equations:

$$C_T \text{ (F g}^{-1}\text{)} = \frac{C_s}{4} \quad (5)$$

where C_s and C_T are the specific capacitances of the electrode and the supercapacitor cell, respectively.

The specific energy density and power density of the supercapacitor cell are defined as follows:

$$E \text{ (W h kg}^{-1}\text{)} = \frac{0.5 C_T \times \Delta V^2}{3.6} \quad (6)$$

$$P \text{ (W kg}^{-1}\text{)} = \frac{E \times 3600}{\Delta t} \quad (7)$$

where ΔV is the potential window during the discharge process after the IR drop, C_T is the gravimetric capacitance of the supercapacitor cell, and Δt (s) is the discharge time. The corresponding gravimetric energy density (E) and power density (P) were obtained from eqn (6) and (7), respectively. The volumetric capacitance, energy density and power density can be obtained by the multiplication of density of active materials and the corresponding gravimetric capacitance, energy density and power density, respectively. m refers to the mass of active materials in the electrode.

3. Results and discussion

Fig. 1a presents a SEM image of flake-structured NOCIC. TEM and HR-TEM images of the NOCIC flakes shown in Fig. 1b and c indicate that the carbon flake contains abundant pores. The EDS spectrum in Fig. 1d confirms the presence of C, N, O and Cl.

The EDS elemental mapping (Fig. S1, ESI†) illustrates that C (red), N (purple), and O (green) are uniformly distributed across the flake. Under low calcination temperatures, the total removal of Cl (blue in Fig. S1, ESI†) is hard, compared to high temperatures.⁹ The SEM image in Fig. 1e shows that the inclusion of transition metal Ti in the synthesis resulted in the decoration of the NOCIC platelet with many nanoparticles. The TEM image in Fig. 1f indicates that some nanoparticles are buried inside the carbon platelets. Further structural details are revealed in the HR-TEM image in Fig. 1g, where these small nanoparticles exhibit well-defined lattice spacings of *ca.* 0.42 nm, 0.27 nm and 0.23 nm, which correspond to $\text{TiN}_{0.98}$, TiO_2 and carbon quantum dots (CQDs).²⁹ The layer-to-layer spacing identified from the HRTEM image (Fig. 1g) is 0.37 nm. The EDS spectrum of the nanoparticle in Fig. 1h shows that these nanoparticles consist of Ti, C, N, O and Cl atoms. The EDS elemental mapping shows that C (red), N (pink), O (blackish green), Ti (green) and Cl (blue) are evenly distributed (Fig. S2, ESI†). Except for the Ti element, the chemical compositions of the carbon flake shown in Fig. 1e are the same as those shown in Fig. 1a. As the content of Ti increases, the Ti based nanoparticles shrink to small-sized and isolated islands on the surface of carbon flakes (Fig. S4, ESI†). These Ti based nanoparticles and CQDs will significantly affect the supercapacitor performance. However, the content of CQDs is little as its signal is out of the detection range in XRD and XPS spectra, making the capacitive contribution from CQDs less important in this study.

The XRD patterns of the as-synthesized NOCIC and TiNOCIC samples are presented in Fig. 2a. The diffraction peaks of the NOCIC, centered at $2\theta = 25^\circ$ and 43.5° , can be attributed to the (002) and (100) reflections of a disordered carbon structure, respectively. The major crystal structures of the TiNOCIC nanocomposites yield peaks at 25° , 37° , 48° and 53° , which can be assigned to the (110), (101), (220) and (204) main phases of anatase TiO_2 , respectively.³⁰ Peaks at 37° , 43° and 62° can be assigned to the (111), (200) and (220) facets of minor phase $\text{TiN}_{0.98}$. Our experiments

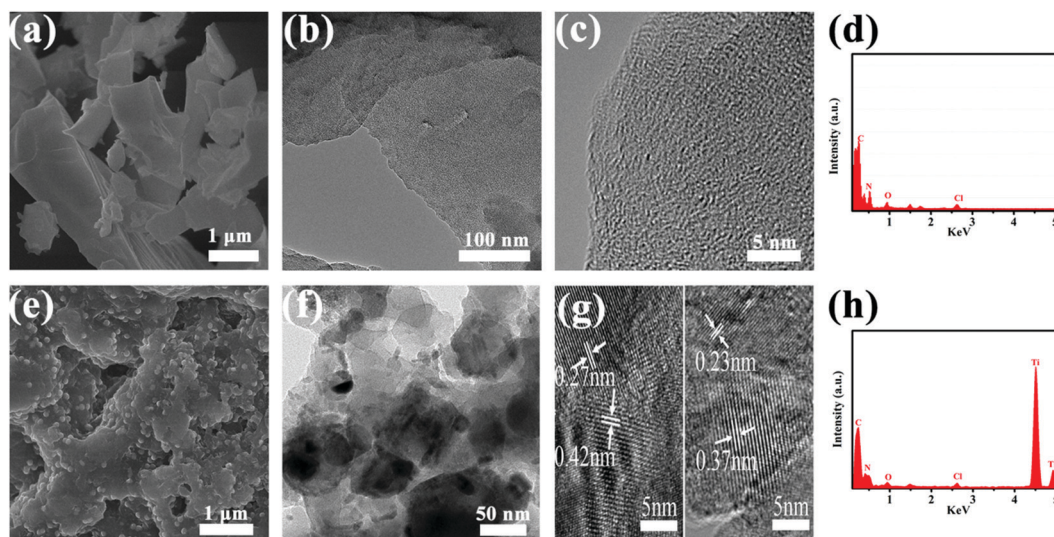


Fig. 1 Characterization of (a–d) NOCIC and (e–h) TiNOCIC. (a and e) SEM, (b and f) TEM, (c and g) HRTEM and (d and h) EDS.

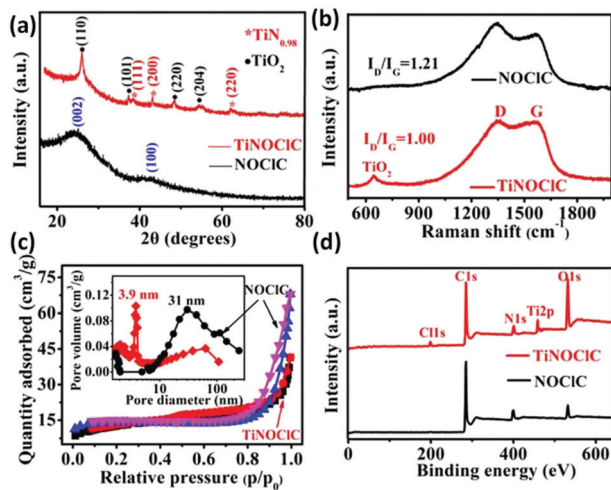


Fig. 2 Structural characterization of NOCIC and TiNOCIC. (a) XRD patterns, (b) Raman spectra, (c) nitrogen adsorption/desorption isotherms, inset: pore size distribution for N_2 , and (d) survey XPS spectra.

show that while increasing the solvothermal reaction temperature improves the crystallinity of the solid products, the content of byproducts such as TiC_2N_8 also increases (Fig. S5, ESI[†]), which has adverse impacts on their electrochemical performance.

Fig. 2b depicts the Raman spectra of NOCIC and TiNOCIC samples. Both types of carbon materials possess two characteristic peaks at 1340 and 1568 cm^{-1} , corresponding to the disordered carbon (D band) and E_{2g} mode phonons of the sp^2 graphitic carbon (G band), respectively. However, the TiNOCIC spectrum has another peak at around 608 cm^{-1} , which is characteristic of the rutile TiO_2 .¹⁸ The intensity I_D/I_G ratio decreased from 1.21 (for NOCIC) to 1.00 (for TiNOCIC), while the area A_D/A_G ratio decreased from 5.16 (for NOCIC) to 2.88 (for TiNOCIC). The reduced I_D/I_G and A_D/A_G ratios can be attributed to the metal doping that annihilated some defective sites in the sp^2 carbon frameworks.^{30–32} The porosity properties of the carbon materials were investigated using nitrogen adsorption-desorption isotherms shown in Fig. 2c, which exhibits the characteristics of hierarchical porosity, including micropores, mesopores and macropores. The hysteresis loop between the adsorption and desorption branches suggests the presence of mesopores. Additionally, the slightly steep adsorption at a relative pressure of 0.8–1.0 is indicative of the presence of macropores.^{33,34} The pore size distributions are presented in the inset of Fig. 2c, where the pore size of approximately 3.9 nm of the mesopores is consistent with that of mesoporous carbon materials. Moreover, much larger pore volumes of mesopores with pore sizes exist in the NOCIC than that in the TiNOCIC. The surface area and pore volume of the NOCIC and TiNOCIC were calculated by the BET and BJH methods, respectively. The BET surface area (S_{BET}) and pore volume of NOCIC decrease after metal doping, from 59.4 $m^2 g^{-1}$ of NOCIC to 44.9 $m^2 g^{-1}$ of TiNOCIC and the pore volume decreased correspondingly, as summarized in Table S1 (ESI[†]).

Furthermore, XPS survey spectra of the NOCIC and TiNOCIC samples in Fig. 2d display the C 1s, N 1s, O 1s, Cl 2p and Ti 2p peaks at the binding energies of 284, 400, 529.6, 198, and

460 eV, respectively. The XPS spectrum suggests that the TiNOCIC sample has a doping content of 18.6 at% for nitrogen, 2.4 at% for oxygen, and 2.9 at% for Ti, respectively. NOCIC has a comparable nitrogen doping content of 16.9 at%. Although XPS only detects the compositions of surface layers, the above result nevertheless suggests that this synthetic method can lead to efficient nitrogen doping.

In the enlarged view of the Ti 2p region in Fig. 3a, four peaks at 457.6, 458.7, 461 and 287.4 eV are observed, which correspond to the Ti–O, and Ti–N–O, respectively.¹⁸ The high-resolution C 1s XPS spectrum in Fig. 3b reveals the oxygen-containing groups C–O at 285 eV and C–N at 285.6 eV.³⁵ The N 1s XPS spectrum can be fitted to three distinct peaks at 398.7, 399.5 and 401.3 eV (Fig. 3c), which originate from the Ti–N, pyridinic N, and quaternary N, respectively. In the O 1s XPS spectrum, two bonds of C–O (532 eV), and TiO_2 (530.2 eV) can be unraveled (Fig. 3d). Both XRD and XPS suggest that TiO_2 is the main product in Ti-based nanoparticles, where most of the Ti sites are on the corners and faces of the cuboid anatase TiO_2 structure, facilitating the exposure of the active sites. High-resolution XPS spectra of NOCIC are presented in Fig. S6 (ESI[†]), where C 1s XPS spectra revealed the oxygen-containing groups at 285 eV (C–O) and 285.6 eV (C–N). The N 1s XPS spectrum can be fitted to two distinct peaks at 399.5 and 401.3 eV, which originate from the pyridinic N and quaternary N, respectively.³⁶ In the O 1s XPS spectrum for NOCIC, two bonds of C–O (532.5 eV) and O–H (530.6 eV) can be unraveled.

The electrochemical capacitive performance of the TiNOCIC sample was first investigated using a three-electrode system in 6.0 M KOH aqueous electrolyte. To highlight the critical role of Ti metal doping, an electrode prepared with a NOCIC sample was also examined as a control under the same conditions. Fig. 4a shows the CV curves of the electrodes prepared with NOCIC and TiNOCIC. The CV curves of TiNOCIC did not show redox peaks, which might be ascribed to the low content of Ti

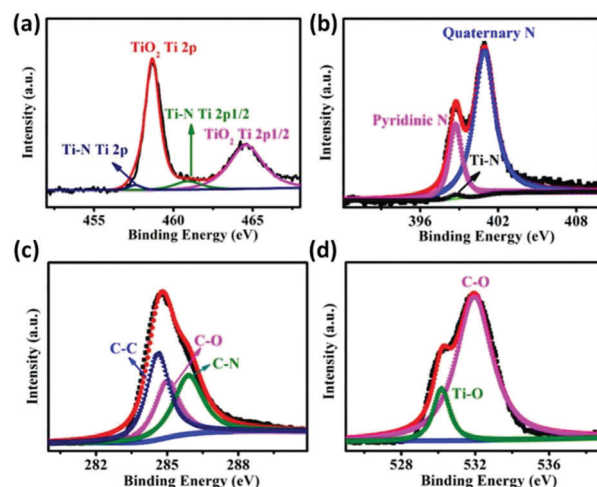
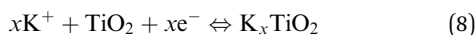


Fig. 3 XPS spectra of TiNOCIC: (a) Ti 2p electrons of TiO_2 and $TiN_{0.98}$, (b) C 1s, (c) N 1s and (d) O 1s. XPS spectrum of the C 1s in (b) shows spin-orbit splitting of electrons of C–C, C–O and C–N. The spectra indicate that Ti–N takes only 5.04% of N content.

and the sluggish redox activity of Ti. The electric double-layer capacitance of NOCIC showed a rectangular shape with a relatively little current. We believe that the true pseudocapacitance is due to the fast and reversible faradaic reactions, which may involve the electrochemical interconversion of metal with different valences in the solid and the intercalation of cations from the electrolyte. Within the small voltage range (due to the thermodynamic limit of water, 1.23 V), the oxidation and reduction of Ti^{4+} ions barely occurred. Thus, the pseudocapacitance in this study is mainly derived from the intercalation of K^+ in and out of the working electrode, such as TiO_2 materials.



The corresponding galvanostatic charge/discharge curves in Fig. 4b are highly symmetric triangles with only a small IR drop even at a current density as high as 80 A g^{-1} , indicating efficient double-layer charge storage, excellent reversibility and high coulombic efficiency. On the other hand, the NOCIC sample exhibited worse rate capabilities (Fig. S7, ESI[†]), revealing the importance of efficient electron exchange on the conductive electrode material with the increased SSA and mesoporous porosity. The obtained capacitance is 48.64 F cm^{-3} at a high current density of 80 A g^{-1} . After the doping of Ti, the TiNOCIC electrode exhibited an outstanding electrochemical performance with higher capacitance (Fig. 4c). A specific capacitance of 285.5 F cm^{-3} was obtained at a current density of 0.5 A g^{-1} .

The as-prepared TiNOCIC electrode exhibits specific capacitances of 220.6, 189.0, 165.6 and 138.0 F cm^{-3} at current densities of 1, 2, 4 and 10 A g^{-1} , respectively, which are much larger than that of NOCIC (Fig. S8, ESI[†]). The above results indicate that Ti has made a significant contribution to the overall capacitance of TiNOCIC. A long cycle life is critical for supercapacitors in practical applications. The cycling stability of the TiNOCIC nanocomposite is demonstrated by repeating the CP test for 40 000 cycles at 30 A g^{-1} (Fig. 4d), in which the initial specific capacitance is retained with nearly no loss after 40 000 cycles. The result suggests that the pore accessibility to electrolyte ions does not change upon the long-term cycling test and highlights the excellent electrochemical stability and outstanding reversibility of the TiNOCIC electrode. The impedance spectra of the NOCIC and TiNOCIC electrodes are presented in Fig. 4e, where the vertical curve in the low-frequency region indicates the nearly ideal capacitive behavior. The intercept at the real axis represents the equivalent series resistance (R_s), which is a combination of resistance of the electrolytes, electronic resistance of the electrode materials and interface resistance. R_s of the TiNOCIC electrode is estimated to be around 1.0Ω . Moreover, τ_0 calculated from the plot of frequency-dependent imaginary capacitance (C') is consistent with that calculated from the bode plot (Fig. 4f). The $>45^\circ$ (negative) phase angle of both TiNOCIC and NOCIC at relatively high frequencies confirms their capacitive behavior at fast rates (Fig. 4c). Specifically, the frequency (of 45°) when the resistance and reactance have equal magnitudes is 148.8 Hz for TiNOCIC, giving a relaxation time ($\tau_0 = 1/f_0$) of 6.7 ms, which indicates

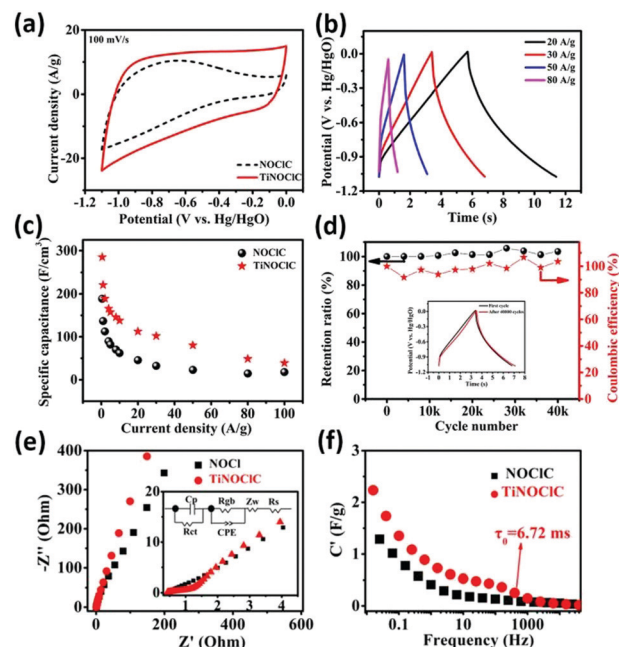


Fig. 4 Electrochemical characterization of TiNOCIC and NOCIC using a three-electrode setup in 6 M KOH electrolyte solution. (a) Cyclic voltammetry at 100 mV s^{-1} for NOCIC and TiNOCIC. (b) Galvanostatic charge/discharge curves upon different current densities from 20 to 80 A g^{-1} . (c) Gravimetric specific capacitances versus current densities. (d) Cycling stability and coulombic efficiency at a current density of 30 A g^{-1} . (e) EIS plots of NOCIC and TiNOCIC. The inset shows the enlarged plot in the high frequency region and the equivalent circuit diagram. (f) Real (C') portions of complex capacitance.

that this material can be charged/discharged under high current density.

In view of practical applications, we used TiNOCIC and NOCIC electrode materials to assemble a symmetrical supercapacitor, and their electrochemical behaviors in a 6 M KOH electrolyte solution were studied in Fig. 5. The CV curves of the symmetrical cell at various scan rates are shown in Fig. 5a, which presents the quasi-rectangle even at a high scan rate of 400 mV s^{-1} , manifesting the superior transport ability of electrons and ions in the cell. Fig. 5b illustrates the GCD curves of the symmetrical cell with the current density in the range of $0.5\text{--}5 \text{ A g}^{-1}$. As seen in Fig. 5c, the TiNOCIC carbon materials exhibit excellent cycle stability, which is far superior to those of the materials currently reported. There is nearly no loss in its specific capacitance after 40 000 cycles at 30 A g^{-1} in the three-electrode configuration compared to the corresponding three-electrode supercapacitors,^{38,39,42,43} and nearly 90% retention after 20 000 cycles at 5 A g^{-1} in the TiNOCIC-based symmetrical supercapacitor (detailed information is provided in Fig. S11, ESI[†]), outperforming most of carbon-based metal-doped two-electrode symmetrical supercapacitors.^{37,40,41} Fig. 5d plots the calculated power density and energy density for the two types of heteroatom doped carbon materials, using the two-electrode system. In general, the power density of TiNOCIC is larger than that of NOCIC. When the current was set at 0.5 A g^{-1} , the energy densities and power densities for TiNOCIC were $11.99/7.92 \text{ W h L}^{-1}$ and

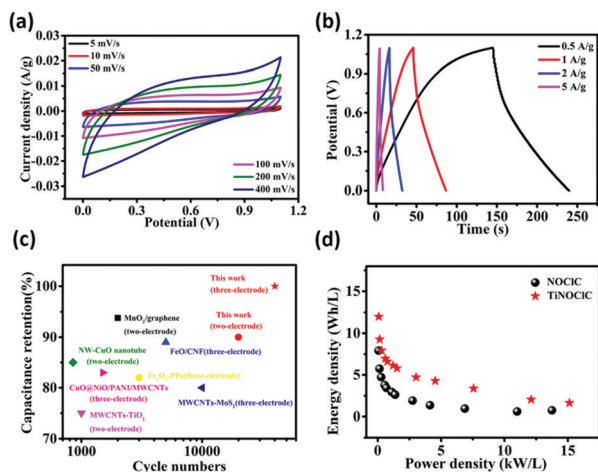


Fig. 5 The electrochemical performance of the TiNOCIC and NOCIC-based symmetrical supercapacitor in 6 M KOH electrolyte solution. (a) CV curves at scan rates of 5–400 mV s^{-1} , (b) GCD curves at current densities of 0.5–5 A g^{-1} , (c) the cycle stability comparison chart between TiNOCIC and other metal-doped materials, such as $\text{MnO}_2/\text{graphene}$,³⁷ $\text{Fe}_2\text{O}_3\text{-PFS}$,³⁸ FeO/CNF ,³⁹ MWCNTs-TiO_2 ,⁴⁰ NW-CuO nanotubes,⁴¹ MWCNTs-MoS_2 ,⁴² and $\text{CuO@NiO/PANI/MWCNTs}$.⁴³ (d) Ragone plots for NOCIC and TiNOCIC.

75.62/68.75 W L^{-1} , respectively. Thus, TiNOCIC is more promising in storing more energy, suggesting that appropriate carbon materials can be used as novel hosts for supporting metal nanoparticles to achieve enhanced energy storage in supercapacitor applications.⁴⁴

4. Conclusions

In summary, this study presents a simple one-step solvothermal method to synthesize Ti, N co-doped carbon materials. This method appears to be very effective in producing high contents of nitrogen doping, which is expected to enhance the specific capacitance due to the pseudocapacitive effect. The inclusion of a small amount of transition metal Ti significantly improved their electrochemical capacitive performance, which may be attributed to the influence of Ti doping on the pore distribution and pseudocapacitance. Under the optimal conditions, the as-prepared TiNOCIC nanocomposite electrode has a high specific volumetric capacitance of 285.5 F cm^{-3} at a current density of 0.5 A g^{-1} , and, impressively, there is nearly no loss in its specific capacitance after 40 000 cycles at a high current density of 30 A g^{-1} . The superior performance of ultrahigh cycling stability and fast charge-discharge capacitance make this TiNOCIC material an attractive candidate for electrochemical energy storage. In addition, the synthetic method presented here can be readily expanded to incorporate other metal elements, fabricating doped carbon/metal oxides/metal composite materials for applications in oxygen reduction reactions, electrochemical gas sensing, etc.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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